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Nano-enhanced cooling techniques for photovoltaic panels: A systematic review and prospect recommendations

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ABSTRACT

The environmental pollution and growing energy demand necessitate the development of power generation using renewable energy systems, especially solar photovoltaic (PV) panels. Despite being an eco-friendly and durable power generation technology, PVs demerit for low conversion efficiency (15–20%). In other meaning, a very low amount of the incident irradiance is converted into electricity, whereas the major is absorbed as heat, uplifting the PVs' surface temperature and declining their efficiency and reliability. Therefore, it is crucial to find a proper cooling technique for upgrading PVs' performance. Recently, along with the nanotechnology development, nano-based PV cooling units have been proposed and investigated as the inclusion of nanoparticles enhances the thermal properties of base materials. Herein, in this paper, the conducted efforts to improve the PVs' performance using nano-based cooing systems were reviewed and discussed. The surveyed PVs' cooling methodologies were classified into separate and hybrid systems: nano-enhanced PCMs, nanofluid-based, and hybrid nano-based cooling units. All relevant equations of materials' properties and PV performance were presented. In addition, the research trend of this field was analyzed by conducting a comprehensive bibliometric analysis using VOSviewer software and based on the Scopus database in the period (2006–2020). Besides, comprehensive discussions of proposed and designed systems were conducted, and major findings were summarized. Based on the literature, using hybrid nano-based cooling systems, the surface temperature reduction can reach 16 °C with enhancement in electrical efficiency up to 50% compared to conventional uncooled PV panels. Furthermore, some future work suggestions were given.

1. Introduction

Currently, renewable energy participates by roughly 13% in global energy consumption, and this portion is neither satisfying nor sufficient for solving energy-related problems. Whereas non-renewable fossil fuels with their environmentally harmful emissions contribute by 81% to worldwide energy consumption (Borges Neto et al., 2010; Hosenuzzaman et al., 2015). Moreover, 2.2% is an annoying and worrying percentage of annual growth of greenhouse gas emissions in the first decade in the current millennium (Change, 2014; Kalaugher, 2014). Hence, under the current global technological and economic widening, which leads to more and more energy demand, putting renewable energy development at the forefront of research priorities is not optional but is mandatory to achieve economic sustainability and

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environmental protection.

There are various types of renewable energy resources that have been developed in utilized on a widespread scale in recent decades. For example, wind energy, which may be utilized for direct driving of sailboats and windmills (Emeis, 2018; Gasch and Twele, 2011; Gipe, 2018) or power generation (Attya et al., 2018; Sharifian et al., 2018; Vargas et al., 2019), and solar energy, which can be utilized for drying operations (Ekechukwu and Norton, 1999; Sharma et al., 2009), cooling and heating systems (Dobriyal et al., 2020; Goel et al., 2020; Qu et al., 2010; Ward et al., 1978), solar still (Elmaadawy et al., 2021; Essa et al., 2020; Peng et al., 2021; Sharshir et al., 2020; Sharshir et al., 2019; Sharshir et al., 2018; Sharshir et al., 2016d), hybrid systems (Abdelgaied et al., 2021; Sharshir et al., 2016a; Sharshir et al., 2016b; Sharshir et al., 2016c), reverse osmosis (Elkadeem et al., 2021; Elmaadawy et al., 2020; Kotb et al., 2021), humidification and dehumidification (Hamed et al., 2015; Kabeel et al., 2014) or for power generation via photovoltaic (PV) panels. There are other resources for renewable energy such as tidal and waves energy (Astariz et al., 2015; Khan et al., 2017; Rusu and Venugopal, 2019), geothermal energy (Ghazvini et al., 2019; Lund and Boyd, 2016; Olabi et al., 2020), and biomass energy (Abbas et al., 2020; Klass, 1998; Perea-Moreno et al., 2019). Furthermore, hybrid renewable energy systems are superb methods to combine more than one renewable resource of energy in one system to increase the obtained benefit from them (Ghorbani et al., 2020; Khare et al., 2016; Sinha and Chandel, 2014).

Depending on the photoelectric effect, sunlight is converted into electricity in PV panels through solar cells made from semiconductor materials (Mickey, 1981; Paranthaman et al., 2016). Solar PV panels provide 16% of the electricity consumed in the world. However, they are also responsible for 20% of the CO_2 emission reduction (Ali, 2020; Energy, 2014). So, PV panels are very important for sustainability and environmental security; hence, their performance and technology must be carefully studied and enhanced. PV panels produce electricity by converting a small amount of incident solar irradiation (15–20%), i.e., they have very low conversion efficiency. At the same time, the rest of incident irradiation is transformed into heat (Teo et al., 2012), resulting in PV panels' surface temperature rise over the ambient by up to 40 °C, which degrades their efficiency (Agrawal and Tiwari, 2010; Kurtz, 2009; Reddy et al., 2015). When PV panel temperature increases only by 1 °C, its efficiency decreases by 0.4–0.65% (Rahman et al., 2015; Shan et al., 2014). Hence, to improve the PV panels' performance, a proper cooling process should be applied.

There are a lot of cooling methods developed for PV panels (Kabeel and Abdelgaied, 2019; Kianifard et al., 2020). Some methods are based on using air (Bambrook and Sproul, 2012; Kandeal et al., 2020; Kumar and Rosen, 2011; Sarhaddi et al., 2010; Shahsavar and Ameri, 2010) or water (Dorobanțu et al., 2013; Moharram et al., 2013; Zhu et al., 2011) as cooling media. Furthermore, using phase change materials (PCMs) (Huang et al., 2011; Kabeel and Abdelgaied, 2019; Velmurugan et al., 2021) and non-water liquids (Han et al., 2013) were proposed and investigated as methods for PV panels' cooling. In addition, applying passive radiative cooling (RC) in PV thermal regulation has been recently studied (Sato and Yamada, 2019; Zhao et al., 2018a). In this methods, various effective coatings can be used; such as anti-reflection coating (Kumar and Chowdhury, 2019), polydimethylsiloxane pyramid-structure (Lee and Luo, 2019), pyramidal micro-structured silica (Zhu et al., 2014), and thin silica micro-gratings (Long et al., 2019), etc. Furthermore, the photonic approach has more recently been paid attention. It merges spectrum selection and RC for the purpose of solar cells' thermal control without any additional systems' complexity or mass. This approach has many benefits: maintaining effective absorption within the solar radiation range, improving the thermal emissivity in IR range, and reflects parasitic heat source (has no participation in the photoelectric conversion) (Fan et al., 2021). Significant temperature reduction has been reported; for example, about 10 °C reduction was obtained via certain optical filter (Sun et al., 2017), and multi-layer optical structure (An et al., 2019). In addition, utilization of photonic structure can result in higher potential cooling energy gain during the day by 124.2% over RC (Zhao et al., 2018b).

Furthermore, Adding nanoparticles to liquids leads to enhancement in their thermal properties and makes them more effective in cooling processes (Bakthavatchalam et al., 2020; Xuan and Li, 2000). So, nanofluids have been applied for PV panels' cooling and showed great enhancement in their efficiency. Moreover, adding nanoparticles to PCM improves their thermal conductivity (He et al., 2012; Leong et al., 2019; Lin and Al-Kayiem, 2016), and they were proposed as PV panel coolants. In this paper, to our best knowledge, the enhancement of PV panel cooling methods by nanoparticles was scoped, analyzed, and summarized to emphasize the importance of nanotechnology and its application in such a vital issue. First, to state the importance of this aspect, the research trend was analyzed via a bibliometric analysis with the aid of a VOSviewer. Then, throughout this manuscript, general equations used for evaluating the performance and obtaining materials' properties were driven. In addition, the conducted nano-based PV cooling investigations were classified into separate or hybrid units (nano-enhanced PCMs (N-PCMs), nanofluids (NFs), and hybrid). All systems' designs and major results were comprehensively discussed. Moreover, recommendations for future works were suggested.

2. Performance evaluation of photovoltaic panels

As known, the PV depends on converting input (solar power) to output (electrical power). Simply, the output power, input power, and electrical (conversion) efficiency can be calculated from Eqs. (1), (2) (Al-Waeli et al., 2017a), and (3) (Rejeb et al., 2016), respectively, as follows:

$$P_{Elec} = I \times V \tag{1}$$

$$P_{Solar} = I_{Solar} \times A_{Panel} \tag{2}$$

$$\eta_{Elec} = P_{Elec}/P_{Solar} \tag{3}$$

where, I is electrical current (A), V is electrical voltage (V), I_{Solar} is solar intensity (W/m^2), and A_{Panel} is the area of a panel that harvests solar radiation (m^2).

It should be noted that the performance of PVs varies with their temperature. Hence, as reported by (Jordehi, 2016), the recommended range of the PV cell temperature is 0–75 °C. The temperature-dependency of output current (Krishan and Suhag, 2019), power, and efficiency (Skoplaki and Palyvos, 2009) of PV cell can be presented as follows:

$$I = I_{PH} - I_o\left[exp\left(\frac{V + (I \times R_S)}{A \times K \times T_{PV}}\right) - 1\right] - \frac{V + (I \times R_S)}{R_{SH}} \tag{4}$$

$$P_{Elec} = P_{PVn}.f_{PV}\left(\frac{I_{Solar}}{I_{Solar,STC}}\right) \times \left[1 + \beta_r\left(T_{PV} - T_{PV,STC}\right)\right] \tag{5}$$

$$\eta_{Elec} = \eta_{T,STC}\left[\gamma.log_{10}(I_{Solar}) + \left(1 - \beta_r\left(T_{PV} - T_{PV,STC}\right)\right)\right] \tag{6}$$

where, I_{ph}, I_o, R_S, A, K, T_{PV}, and R_{SH} in the 4th equation are inverse saturation current (A), photo-current (A), series resistance (Ω), junction material factor (-), Boltzmann constant (1.3805×10^{-23} J/K), actual PV cell temperature (K), and shunt resistance (Ω), respectively. On the other hand, P_{PVn}, f_{PV}, $I_{Solar,STC}$, β_r, $T_{PV,STC}$, and $\eta_{T,STC}$ are PV array's output nominal power (W), derating factor (%), received hourly solar radiation at standard test conditions (STC = 25 °C), PV cell efficiency/temperature coefficient (%/°C) (0.4–0.5%/ °C) (Hove, 2000), cell temperature at STC (K), and efficiency at STC (%). Moreover, the radiation-intensity coefficient for cell efficiency (γ) is generally assumed = zero (Dubey et al., 2013; Evans, 1981).

For PV/T (photovoltaic/thermal) systems, the extracted heat from the panel cooling process, by a working fluid (liquid or air), can be used

in other applications, hence, there are two useful outputs: electrical and thermal energies. The useful thermal energy can be calculated by using Eq. (7) (Hjerrild et al., 2016), and the thermal efficiency can be determined from Eq. (8) (Al-Waeli et al., 2017a). Accordingly, the overall system efficiency can be evaluated by adding electrical and thermal efficiencies, as illustrated in Eq. (9) (Yazdanifard et al., 2017).

$$Q_{Utilized} = \dot{m}C_P(T_{Outlet} - T_{Inlet}) \tag{7}$$

$$\eta_{Thermal} = Q_{Utilized}/P_{Solar} \tag{8}$$

$$\eta_{Overall} = \eta_{Elec} + \eta_{Thermal} = (P + Q_{Utilized})/P_{Solar} \tag{9}$$

where $\dot{m}$ is the mass flow rate of working fluid (kg/s), and $(T_{Outlet} - T_{Inlet})$ is the temperature difference attained by the working fluid through the thermal system (K).

3. Overview of nano-enhanced PV cooling techniques

3.1. Classification of nano-enhanced PV cooling techniques

Generally, the PV cooling methodologies have been classified in different ways. According to (Kandeal et al., 2020), these techniques can be classified according to heat transfer process: convective (via liquid or air), conductive (via PCMs), and radiative (via selective emissive coatings) cooling. Based on this classification, the nanomaterials can be integrated into liquid and PCM-based PV cooling. Accordingly, nano-based PV cooling can be classified into nanofluid (NF), nano-enhanced PCM (N-PCM), and hybrid (NF + N-PCM). On the one hand, the N-PCM unit is utilized for stand-alone PV cooling either at direct or indirect contact with PV modules. On the other hand, both NF and hybrid (NF + N-PCM)-based units are used in PV/T systems. Also, in hybrid nano-based PV/T systems, the N-PCM unit can be directly contacted to the PV module or used in separate reservoirs. This classification was considered throughout this paper, and corresponding works were detailly discussed in Sections 3–5.

3.2. Research trend of nano-enhanced PV cooling techniques

The field of nano-based PV panels' cooling is a promising research work that has attracted researchers' attention. To investigate the extent of this trend, VOSviewer software was used with the aid of the Scopus database. This database was selected as it is considered one of the most known and oldest databases. The used keywords were "PV", "PV/T", "PCM", "nano-enhanced" and "nanofluid." The results were restricted by excluding review articles and defining the period between 2016 and 2020. The common keywords between the obtained results were classified into seven clusters as shown in the network visualization map illustrated in Fig. 1. The red cluster contains some items such as photovoltaic, PCM-cooling, PV/cooling, latent heat, and temperature regulation. The strong links indicate the focus of publications in this cluster, and the strongest link in this cluster is the link between photovoltaic and PCM-cooling. According to the reviewed studies, examples for the studies that scooped PCM-cooling of PV panels are (Jamil et al., 2021; Nada et al., 2018; Salem et al., 2019; Sarafraz et al., 2019; Sharma et al., 2017). Meanwhile, the green cluster contains items: nanofluid, renewable energy, hybrid PV/T, heat pump, and entropy generation. While the purple cluster contains items: PV/T, optical properties, and overall efficiency. The strongest link in the last two clusters is the link between PV/T and nanofluid. Examples from the reviewed references that scoop this strong link are (Al-Waeli et al., 2017b; Das et al., 2020; Hussain and Kim, 2018; Sardarabadi et al., 2017; Sathyamurthy et al., 2021). Additionally, the blue cluster contains several items such as electrical efficiency, thermal efficiency, nanoparticles, heat pipe, and optical filter. The most definite link in this cluster is between electrical efficiency, thermal efficiency, and photovoltaic, which indicates the reviewed references carried out the efficiency calculations such as (Abdallah et al., 2019; Abdelrazik et al., 2020; Abdollahi and Rahimi, 2020; Al-Waeli et al., 2017b; Alous et al., 2019). The other three clusters (light blue, orange and yellow) have small dimensions and weak links with other clusters. The focus of research trends in the period of 2016–2020 is shown in Fig. 2. As shown, the red cluster indicates the

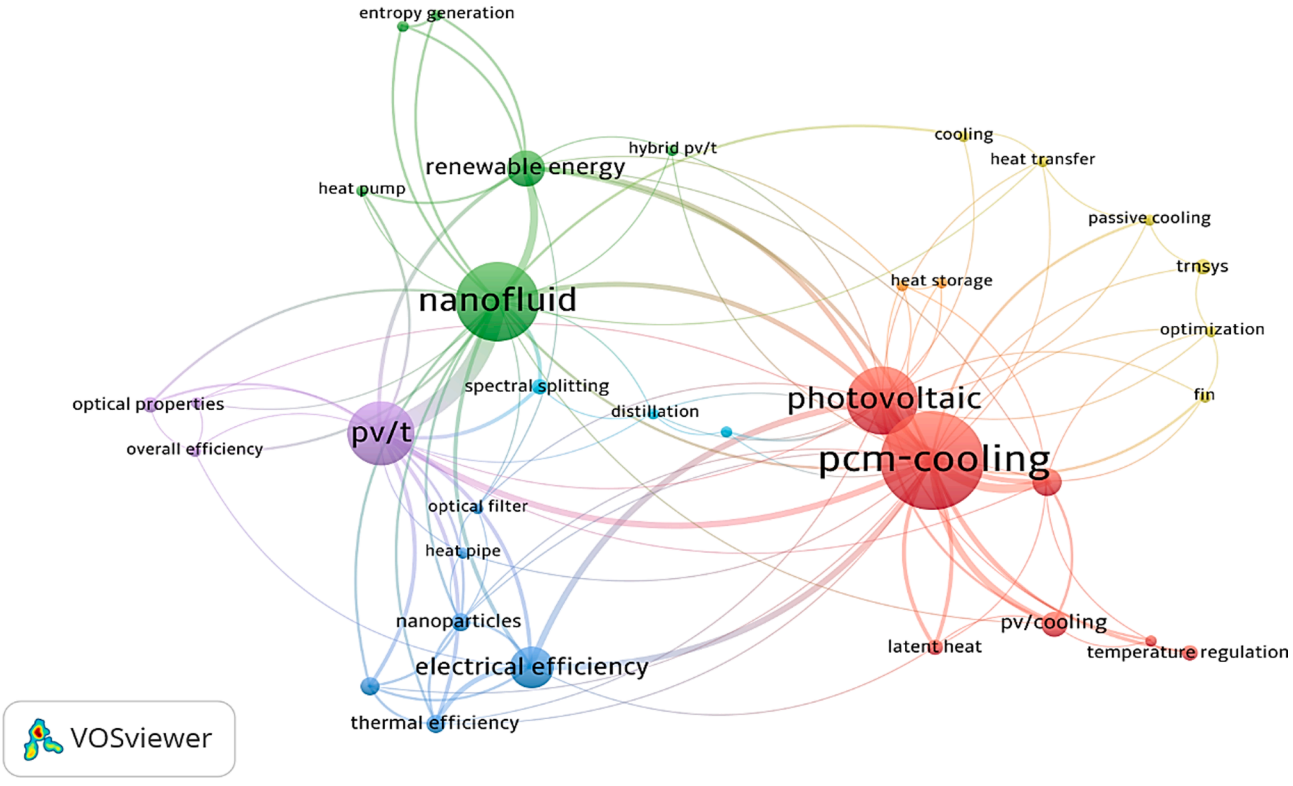


Fig. 1. Network visualization map.

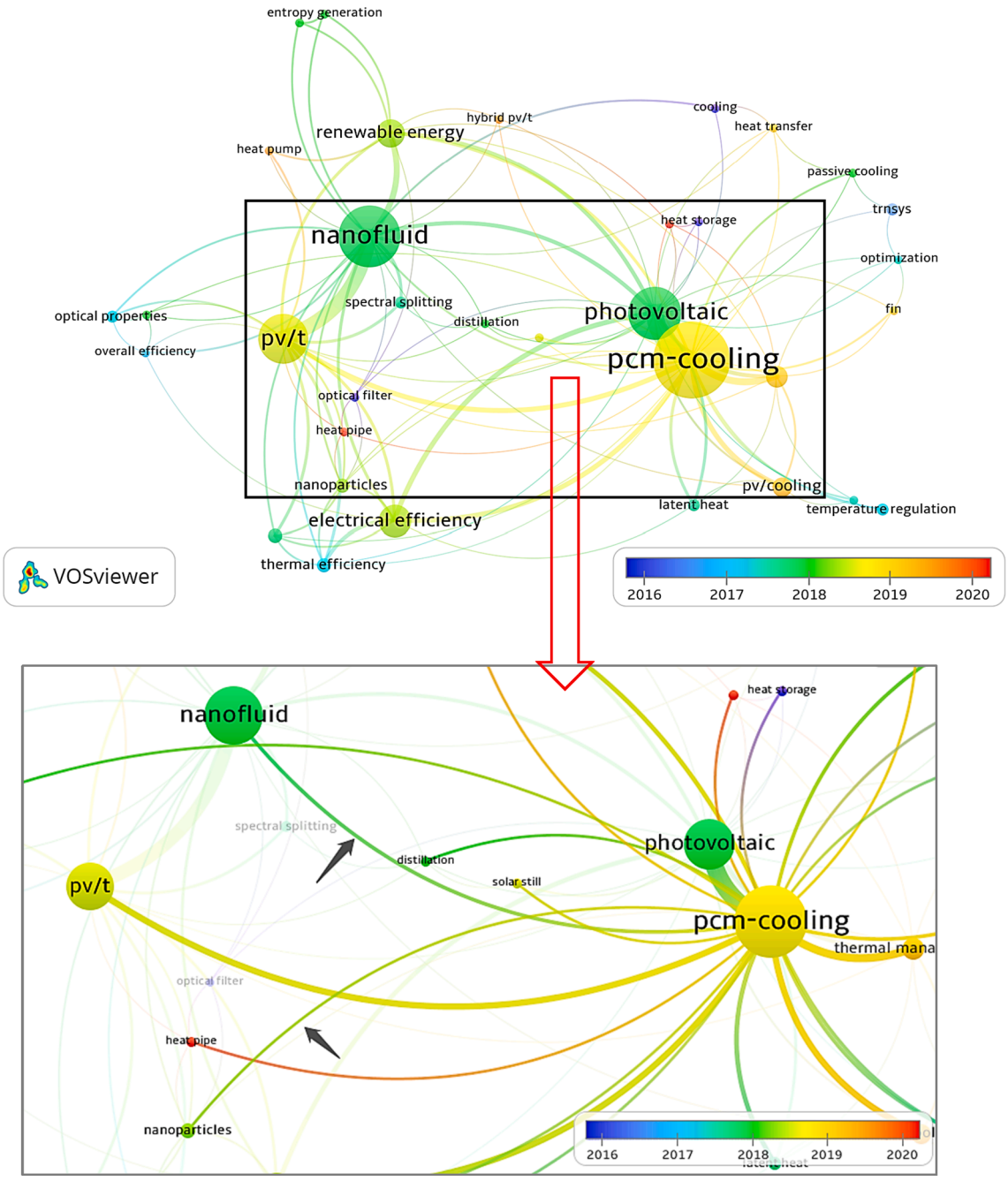


Fig. 2. Focus of research publications in the period between 2016 and 2020.

publications focus on 2020, while the dark and light blue colors indicate the publications focus in 2016 and 2017, respectively. Meanwhile, the green cluster indicates the publications' focus in 2018, which includes nanofluid applications with photovoltaic panels. Furthermore, the yellow cluster indicates the publications' focus in 2019, which includes integrating PCM-cooling with PV/T systems. This analysis proves the importance of nano-based PV cooling via N-PCMs, NFs, and hybrid systems.

4. PV cooling via nano-enhanced phase-change materials (N-PCM)

Generally, phase change materials (PCMs) are used to store thermal energy by gaining a large amount of latent heat during their phase transition period (charging process), and further releasing it in discharging process. Unfortunately, the low thermal conductivity of PCMs limits their effectiveness in thermal applications. So, nanoparticles with good thermal conductivity can be added to PCMs to form highly thermally conductive N-PCMs. Both pure and N-PCMs have been widely used in PV panels' cooling (Siecker et al., 2017), but herein in the next sections, the latter is focused. For systems' modeling and analysis, thermal and rheological properties of pure or N- PCMs should be known. These properties can be determined by experiments or given by suppliers. For the N-PCMs, properties can be determined by experiments or calculated depending on base materials properties.

4.1. Properties evaluation of N-PCMs

4.1.1. Rheological properties

The most important rheological property for heat storage material is

its density. For pure PCM, the density changes according to the phase, depending on its temperature concerning average, first, and second melting temperatures of the material. The density of PCM at different phases can be calculated from (Abdelrazik et al., 2020) as follows:

$$\rho_{PCM}=\begin{cases}\rho_{PCM-Solid} & ,T_{PCM}\leqslant T_{PCM-M1}\\ f_{Liquid}\rho_{PCM-Liquid}+\left(1-f_{Liquid}\right)\rho_{PCM-Solid} & ,T_{PCM-M1}<T_{PCM}\leqslant T_{PCM-M}\\ \rho_{PCM-Liquid} & ,T_{PCM}>T_{PCM-M2}\end{cases} \tag{10}$$

where T_{PCM-M1} and T_{PCM-M2} are first and second melting temperatures (K). While T_{PCM-M} is the average melting temperature (K) and equals $\frac{1}{2}(T_{PCM-M1}+T_{PCM-M2})$. Besides, f_{Liquid} is PCM liquid fraction that can be calculated (Abdelrazik et al., 2020) as:

$$f_{Liquid}=\frac{T_{PCM}-T_{PCM-M1}}{T_{PCM-M2}-T_{PCM-M1}} \tag{11}$$

On the other hand, N-PCM is considered a mixture of PCM and nanoparticles, so its density can be calculated by (Maxwell, 1881):

$$\rho_{nano-PCM}=\phi_{nano}\rho_{PCM-Liquid}+(1-\phi_{nano})\rho_{PCM-Solid} \tag{12}$$

where $\rho_{PCM-Solid}$ and $\rho_{PCM-Liquid}$ are the density of PCM at solid and liquid states, respectively (kg/m^3), and ϕ_{nano} is the fraction of volumetric nanoparticles concentration.

4.1.2. Thermal properties

Herein in this part, the presented mathematical relations are used for evaluating the thermal properties of PCMs: thermal conductivity, latent heat, and specific heat.

Firstly, the thermal conductivity of pure and N-PCMs at different phases can be obtained employing Eq. (13) (Abdelrazik et al., 2020), and Eq. (14) (Karunamurthy et al., 2012; Sharma et al., 2017), respectively, as follows:

$$k_{PCM}=\begin{cases}k_{PCM-Solid} & ,T_{PCM}\leqslant T_{PCM-M1}\\ f_{Liquid}k_{PCM-Liquid}+\left(1-f_{Liquid}\right)k_{PCM-Solid} & ,T_{PCM-M1}<T_{PCM}\leqslant T_{PCM-M}\\ k_{PCM-Liquid} & ,T_{PCM}>T_{PCM-M2}\end{cases} \tag{13}$$

$$k_{nano-PCM}=\left(\frac{k_{nano}+2k_{PCM}-2(k_{PCM}-k_{nano})\phi_{nano}}{k_{nano}+2k_{PCM}+2(k_{PCM}-k_{nano})\phi_{nano}}\right)k_{PCM} \tag{14}$$

where $k_{PCM-Solid}$ and $k_{PCM-Liquid}$ are thermal conductivity of PCM at solid and liquid states, respectively (W/m K).

Secondly, latent heat of pure materials is known for the specified applied material. Whereas latent heat of N-PCM can be evaluated by applying Eq. (15) (Bondareva et al., 2019).

$$h_{l_{nano-PCM}}=(1-\phi_{nano})\rho_{PCM}h_{l_{PCM}}/\rho_{nano-PCM} \tag{15}$$

Finally, Eq. (16) (Atkin and Farid, 2015), Eq. (17) (Maxwell, 1881) are used to calculate specific heat for pure and N-PCM at different phases (Atkin and Farid, 2015).

$$C_{P_{PCM}}=\begin{cases}C_{P_{PCM-Solid}} & ,T_{PCM}\leqslant T_{PCM-M1}\\ \dfrac{4h_{l_{PCM}}-4C_{P_{PCM-Solid}}(T_{PCM-M2}-T_{PCM-M1})}{(T_{PCM-M2}-T_{PCM-M1})^2}(T_{PCM}-T_{PCM-M1})+C_{P_{PCM-Solid}} & ,T_{PCM-M1}<T_{PCM}\leqslant T_{PCM-M}\\ \dfrac{4h_{l_{PCM}}-4C_{P_{PCM-Average}}(T_{PCM-M2}-T_{PCM-M1})}{(T_{PCM-M2}-T_{PCM-M1})^2}(T_{PCM}-T_{PCM-M1})+C_{P_{PCM-Average}} & ,T_{PCM-M}<T_{PCM}\leqslant T_{PCM-M2}\\ C_{P_{PCM-Liquid}} & ,T_{PCM}>T_{PCM-M2}\end{cases} \tag{16}$$

$$C_{P_{nano-PCM}}=\frac{\phi_{nano}\rho_{nano}C_{P_{nano}}+(1-\phi_{nano})\rho_{PCM}C_{P_{PCM}}}{\rho_{nano-PCM}} \tag{17}$$

where $C_{P_{PCM-Solid}}$ and $C_{P_{PCM-Liquid}}$ are specific heat of PCM at solid and liquid states, respectively (J/kg K). While $C_{P_{PCM-Average}}$ is average specific heat for PCM (J/kg K) which equals $\frac{1}{2}\left(C_{P_{PCM-Solid}}+C_{P_{PCM-Liquid}}\right)$.

4.2. Improved performance of PV panels cooled via N-PCMs

Many investigations concerning applying N- PCMs in PV cooling have been conducted to present the enhanced performance. (Sharma et al., 2017) experimentally investigated the influence of using micro-fins, pure, and N-PCMs for PV panels cooling integrated into buildings. For real application miming, an aluminum plate was located on the PCM container, which was insulated, and a heater was installed over this plate. The experiments were carried out for aluminum plate with and without micro-fins and for pure and N-PCMs. The used PCM was commercial paraffin wax because it is cheap, eco-friendly, non-toxic, available, and advantageous with high fusion latent heat. Copper oxide (CuO) was selected as a nanomaterial because of its availability, relatively low cost, chemical indolence with paraffin wax, and high thermal conductivity. Compared to the basic case of natural convection, the results revealed that applying pure and N-PCMs with un-finned plate led to a decrease in panel temperature by 9.6 and 11.2 °C, respectively. Subsequently, compared to using micro-fins only, the results showed that applying micro-fins with pure and N-PCMs led to a reduction in panel temperature by 10.7 and 12.5 °C, respectively. It has been noticed that integrating PV panels in building facades causes insulation for the panel, hence their temperature increase, and the generated electricity decreases. (Nada et al., 2018) experimentally investigated applying N-PCM to overcome this problem. Three PV modules were fixed on a frame for conducting an outdoor experiment under the meteorological conditions of Giza, Egypt. The first module was a PV panel with 2 Inches thickness wood plate on its backplate surface to simulate the thermal insulation of building facades. The second module was PV panel with a container of PCM (pure RT_{55} paraffin wax) on its backplate surface. While the third one was the same as the second one, but N-PCM with Al_2O_3 (2%) nanoparticles was used. The temperature regulation, efficiency, and performance of the second and third modules were much better than the first one. The results revealed that applying pure and N-PCM to the back surface of the PV panel increased its efficiency by 5.7 and 13.2% and reduced surface temperature by 8.1 and 10.6 °C, respectively. (Abdelrahman et al., 2019) experimentally studied how heat sink configurations, using pure or N-PCM, and changing the applied heat flux could be able to positively affect the performance of the front surface of PV panels. To simulate PV panel and solar radiation stored in it, a voltage regulator was connected to a heating element which was located between upper and lower aluminum plates. To enhance the thermal conductivity of the lower plate, a heat sink (consists of a copper sheet with 1

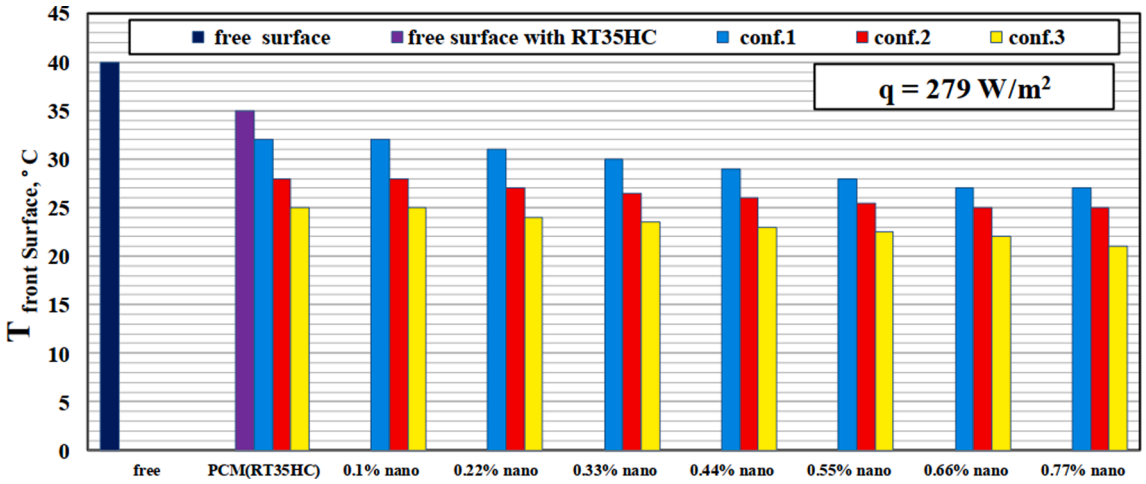


Fig. 3. Variation of PV surface temperature using finned heat sink immersed in N-PCM with various nano concentrations, at heat flux of 279 W/m^2 (configurations 1, 2 and 3 are heat sink with 11, 18 and 22 fins, respectively) (Abdelrahman et al., 2019).

mm thickness and hollow copper cylinders soldered on it as fins) was attached to it. The heat sink was immersed in PCM, which was contained inside a Perspex box. A primary experiment was carried out by applying the same conditions described by (Huang et al., 2006) on the simulation setup and the results showed a good agreement which proved the validation of simulation setup results. Several experiments were carried out under different conditions: heat flux (820, 514, and 279 W/m^2), heat sink configuration (without, with 11, with18, and with 22 fins), and PCM. The applied PCM conditions were as following: without PCM, with RT35HC PCM, and with nano-Al_2O_3-enhanced PCM at different concentrations (0.1, 0.22, 0.33, 0.44, 0.55, 0.66 and 0.77 vol%). For instance, Fig. 3 illustrates the measured PV surface temperatures resulted using 279 W/m^2 for all cases of heat sink and PCM. The results showed that using a heat sink with fins and nano-Al_2O_3-enhanced PCM introduced a good solution for the reduction of PV panel surface temperature by percentage ranges between 36.9 and 52.3%. The nanoparticles concentration in N-PCM played a great role in surface temperature reduction. Changing the concentration from 0.1 to 0.77 vol % caused 1–4.5 °C reduction in the surface temperature.

(Jamil et al., 2021) experimentally tested the performance of cooled PV panel with pure and N-PCM (PT-58) compared to uncooled ones. The experiments were carried out to figure out the effect of changing nanoparticles concentration (0.25 and 0.5 wt%) and type (multi-walled carbon nanotubes (MWCNT), graphene nanoplatelets (GNP), and magnesium nanoparticles) on PV panel temperature, electrical efficiency, and electrical power output. The best results were observed by adding 0.5 wt% graphene nanoplatelets to PCM, which decreased the PV panel temperature by 9.94 °C, enhanced the electrical efficiency by 11.11%, and increased the electrical power output by 33.07%.

5. PV panels' cooling via nanofluids (NFs)

5.1. Evaluation of NF properties

In the following part, the equations used to calculate the NF properties depending on the base fluid and nanoparticles properties are driven.

For the thermal conductivity, it can be evaluated by using the following equation (Maxwell, 1873):

$$k_{nanofluid} = \frac{\left[\left(k_{nanoparticles} + k_{base-fluid}\right) + 2\phi + \left(k_{nanoparticles} - k_{base-fluid}\right)\right]}{\left[\left(k_{nanoparticles} + 2k_{base-fluid}\right) - \phi\left(k_{nanoparticles} - k_{base-fluid}\right)\right]} \quad (18)$$

where $k_{nanofluid}$, $k_{nanoparticles}$, and $k_{base-fluid}$ are thermal conductivities of NF, nanoparticles, and base fluid, respectively (W/m K).

The specific heat of NF can be evaluated by the aid of the following relation (Drew and Passman, 2006):

$$C_{nanofluid} = C_{nanoparticles} + C_{base-fluid}(1-\phi) \quad (19)$$

where $C_{nanofluid}$, $C_{nanoparticles}$ and $C_{base-fluid}$ are specific heats of NF, nanoparticles, and base fluid, respectively (J/kg K).

For the NF density, the following equation can be applied (Drew and Passman, 2006; Hussain and Kim, 2018):

$$\rho_{nanofluid} = \rho_{nanoparticles}(\phi) + \rho_{base-fluid}(1-\phi) \quad (20)$$

where $\rho_{nanofluid}$, $\rho_{nanoparticles}$ and $\rho_{base-fluid}$ are the density of NF, nanoparticles, and base fluid, respectively (kg/m^3).

Finally, the following equation can be applied to calculate the dynamic viscosity of the NF (Brinkman, 1952):

$$\mu_{nanofluid} = \frac{\mu_{base-fluid}}{(1-\phi)^{2.5}} \quad (21)$$

where $\mu_{nanofluid}$, and $\mu_{base-fluid}$ are the dynamic viscosity of NF and base fluid, respectively (kg/m).

5.2. Improved performance of PV panels via NF-based cooling

One of the cooling methods is using a coolant (fluid) to absorb heat from PV panels to be used, in most, in another thermal application, and these systems are called, as previously mentioned, photovoltaic/thermal (PV/T) systems. Adding nanomaterial with high thermal conductivity to the coolant helps in reducing the PV panel temperature and enhances the generated electrical power. Herein in this section, the effect of NF-based cooling on the performance of the PV panel is mainly presented. Many NF-based cooling techniques applied to PV panels have been investigated (Said et al., 2018). The most common techniques depend on circulating the NFs through channels under PV panel and then cooling it through a heat exchanger to benefit from this thermal energy. Many studies were carried out to figure out the affecting parameters on this cooling technique, such as NF type, nanoparticles concentration, NF flowrate, and cooling channels configuration. These related studies were conducted either experimentally (Al-Shamani et al., 2016; Cui and Zhu, 2012; Ghadiri et al., 2015; Michael and Iniyan, 2015) or numerically (Khanjari et al., 2016; Rejeb et al., 2016; Xu and Kleinstreuer, 2014). Furthermore, the NF-based cooling technique has been applied to PV panels integrated with solar radiation concentration methods (An et al., 2016; Radwan et al., 2016; Xu et al., 2014) such as Fresnel lens, reflectors, quartz tubes, and spectral beam splitters. Moreover, the NF-

coolant itself can be used as a solar radiation concentrator (Jing et al., 2015; Saroha et al., 2015). On the other hand, another cooling technique depending on flowing the NF over the PV panel was experimentally studied by Tang and Zhu (Tang and Zhu, 2014).

Recently, (Sathyamurthy et al., 2021) carried out an experimental study to figure out the effect of cooling PV panels by NF and pure water compared to the conventional PV panel. A spiral copper tube was installed in the back surface of the PV panel to provide the path for cooling fluid which absorbs the heat from the panel and releases this thermal energy in a refreshment heat exchanger. The tested cooling fluids were pure water and 2% (1:1) carbon nanotubes/Al_2O_3-water hybrid NF. The findings showed that the average temperature of the hybrid NF-cooled PV panel was lower than water-cooled and conventional PV panels by 2 and 13.5 °C, respectively, with a higher electrical power output of 9.02 and 21.73%. The electrical efficiency of conventional, water-cooled, and hybrid NF-cooled PV panels are 14.8, 16.05, and 17.21%, respectively. In addition, a PV/T collector performance was investigated with three coolants: 0.5 wt% MWCNT-water nanofluid, 0.5 wt% GNP-water nanofluid, and distilled water by (Alous et al., 2019). The results were evaluated from both energetic and exergetic points of view. From the energetic viewpoint, the MWCNT-water nanofluid showed high performance for energetic conversion, but the highest thermal energetic efficiency was released from GNP-water NF. By combining the PV module with the thermal unit, the total energetic effectiveness increased by 57.2% for MWCNT-water nanofluid, 53.4% for distilled water, and 63.1% for GNP-water nanofluid, while the total exergetic efficiency improved by 12.1, 11.2, and 20.6%, respectively. (Abdallah et al., 2019) carried out experimental outdoor tests for PV/T system depending on MWCNT-water NF (flowrate = 1.2 L/min, and concentration range = 0–0.3 vol%) as heat absorption agent. The results confirmed that mixing nanoparticles with the base fluid leads to a significant increment in the thermal and electrical efficiency of the PV/T system. The best efficiencies for the PV/T system were achieved at a nanoparticles concentration of 0.075%. At this concentration, the overall system efficiencies at noon and daytime (average) were 83.26 and 61.23%, respectively. (Hussain and Kim, 2018) mathematically investigated the effect of using pure water and water-based NFs (Al_2O_3, and CuO) as cooling agents on PV panels' technical, economic, and environmental performances. The cooling fluid was assumed to flow equally in parallel copper tubes installed in the back of the PV panel with direct contact with the absorber plate and surrounded by insulation material. The enhanced thermal properties of Al_2O_3 and CuO-based NFs, compared to pure water, led to increment in electrical power output and decrement in electrical energy cost by (36.64 and 74%), and (62.9 and 80%), respectively. Furthermore, considering 30 years as a lifetime of the PV/T system, the economic analysis showed that the payback period of systems with Al_2O_3 and CuO NFs was lower than the system with pure water by (0.5–0.7) and (0.9–1.2) years, respectively. The environmental analysis illustrated that NF-cooled PV panels are effective and should be used in typical applications instead of water-cooled PV panels. (Das et al., 2020) numerically studied the superiority of NF-based cooling of PV panels compared to water-cooling. The tested NFs were MXene (Ti_3C_2)-water + ionic liquid (IL), MXene (Ti_3C_2)-palm oil, and Al_2O_3-water. The nanoparticles' concentration for all tested NFs was 0.2 wt%. The cooling fluid tubes were assumed to be installed in the back of PV panel with a serpentine configuration. The results revealed that cooling PV panels by MXene (Ti_3C_2)-water + (IL), MXene (Ti_3C_2)-palm oil, and Al_2O_3-water NFs could reduce the PV panel temperature by 12.5, 10.7, and 6.9 °C, respectively, compared to the water-cooled PV panel. PV panel cooled by MXene (Ti_3C_2)-water + (IL) NF showed the highest electrical efficiency of 12.2 and 13.95% at 0.01 and 0.07 kg/s flowrates, respectively. With the aid of mathematical modeling and CFD simulation, (Hussain et al., 2019) investigated the effect of the dual-fluid cooling (air + NF) technique on PV panel performance. As shown in Fig. 4, the NF coolant path was parallel copper tubes and directly adhered to the back surface of the PV panel. Furthermore, the cooling air was circulated in air duct equipped by staggered baffles to increase the heat transfer area. The investigated NFs were SiO_2-water, Al_2O_3-water and CuO-water with nanoparticles concentration ranged between 0 and 0.75%. The PV/T system worked by 0.75% CuO-water NF and air had the best performance and showed total efficiency of 90.3%, while the PV/T system worked by pure water and air showed total efficiency of 79.8%.

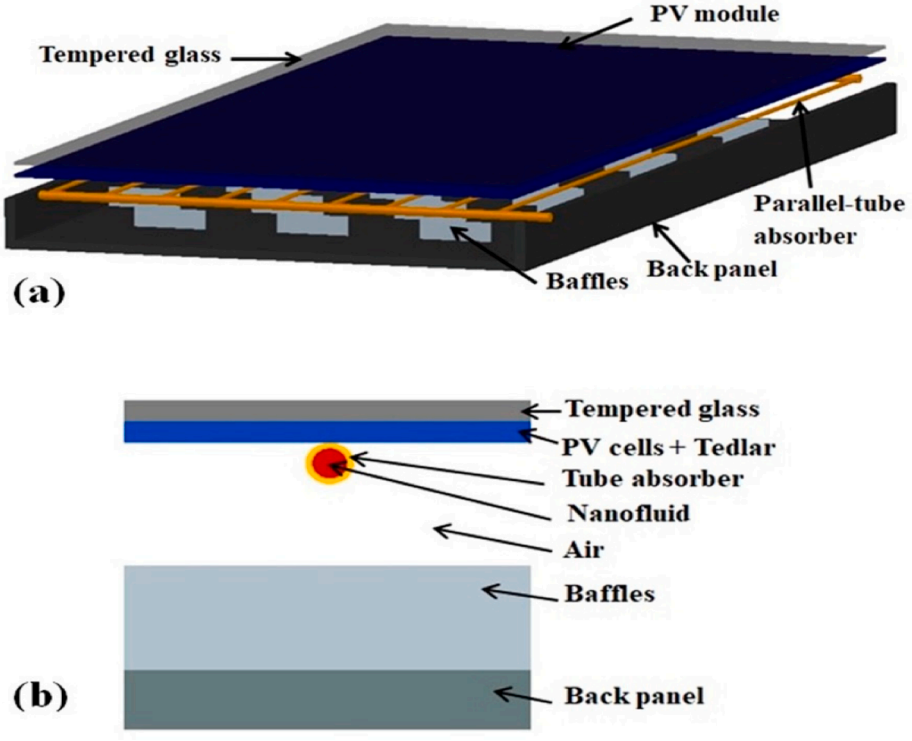
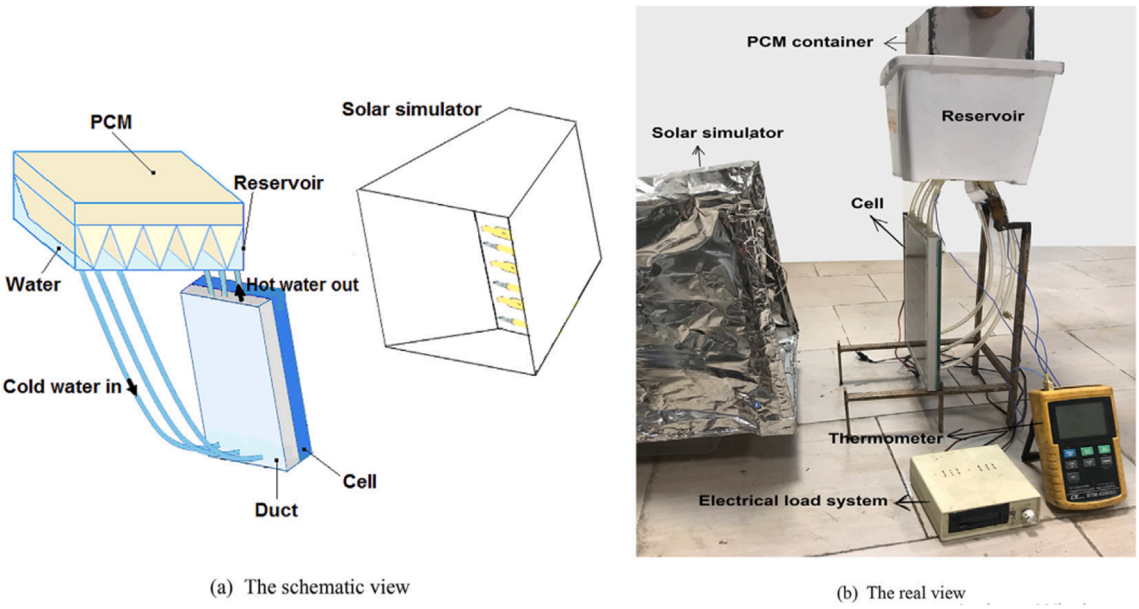


Fig. 4. Exploded view for dual-fluid cooling system (a) isometric view (b) cross-sectional view (Hussain et al., 2019).



(a) The schematic view (b) The real view

Fig. 5. An indoor experiment for PV cooling via flowing water refreshed in N-PCM (Abdollahi and Rahimi, 2020).

6. Hybrid nano-based cooling of PV panels

Based on the previously reviewed works, PV panels can be cooled either using working fluid (on upper or back surfaces), or PCM. The former (working fluid) technique depends on sensible heat extraction that may result in non-uniform temperature distribution on the panel surface. Whereas the latter (PCM) technique depends on latent heat extraction that may have low heat rejection ability. Hence, the hybrid cooling system for PV panels that combines cooling by fluid flow and PCM can be effectively utilized to avoid the two previous problems. Cooling PV panels by water and refreshing this water with nano-powder-enhanced PCM is a good idea to enhance the efficiency of PV panels that were presented and experimentally studied by (Abdollahi and Rahimi, 2020) using the system shown in Fig. 5. A duct with three inlet and outlet ports was attached to the back of PV panel to create a passageway for cooling water to subtract the heat accumulated in the PV panel due to exposing to a solar simulator. Then, the hot water naturally flowed upward to enter the container of PCM to be refreshed, and finally, the water entered the cooling duct again by gravity. So, the water circulation process was carried out without pumps or energy consumption. It was desired to conduct the experiments with PCM owing melting temperature of 25 °C but the purchased coconut oil had a 29 °C melting point, so the used PCM (melting temperature of 25 °C) was a mixture composed of 82 wt% coconut oil and 18 wt% sunflower oil. A 0.009 w/w Boehmite nano-powder which has high thermal conductivity, was mixed with PCM to enhance the system cooling efficiency. To enhance the heat transfer process between PCM and cooling water, PCM container was manufactured in a zigzag shape to increase the contact area between them. The results revealed that the proposed hybrid cooling system enhances the efficiency of PV panels under 410, 530, and 690 W/m^2 radiation intensities by 44.74, 46.3, and 48.23%, respectively, compared to the uncooled PV panel.

(Nasef et al., 2019) carried out a theoretical investigation on the performance of a concentrated PV panel with direct and indirect PCM cooling methods. For direct PCM cooling, pure RT35HC PCM was directly integrated into the backside of a concentrated PV panel. For indirect PCM cooling, the backside of the concentrated PV panel was cooled by circulated cooling fluid in a duct; then the cooling fluid was refreshed by PCM in an aluminum tank. Hence, the second cooling method can be considered as the hybrid cooling method; the results showed that using the hybrid cooling method reduced the concentrated PV panel temperature by 60% compared to the direct PCM and only water-cooling methods. Furthermore, replacing pure water with 4% CuO-water NF as cooling fluid reduces the concentrated PV panel temperature by 4 °C. Subsequently, the efficiency enhances by 2.7%.

For direct contact of the hybrid cooling system with the PV panel, (Salem et al., 2019) presented an experimental investigation for the effect of combining and separating water flow and N-PCM cooling techniques on PV panels' performance. The experiments were carried out under different conditions for water flowrate (0–1 L/min) and (0–1 wt%) Al_2O_3 nanoparticles concentration in PCM ($CaCl_2H_{12}O_6$). The experimental setup consisted of two PV panels: the reference one and another one integrated with 20 square aluminum tubes in its back. These tubes were used for containing cooling water and N-PCM; the tested occupation ratios of PCM were 0, 25, 50, 75, and 100% (0, 5, 10, 15, and 20 tubes). With rough consideration for accomplished PV panel temperature reduction, it can be noted that changing nanoparticles concentration in PCM from 0 to 1 wt% reduces PV panel temperature by ~ 4.5 °C in all experimented cases. The performance of the tested PV panels was evaluated by calculating electrical, thermal, and overall exergy efficiencies. The results illustrated that filling 25% of the tubes by 1 wt% N-PCM and using cooling water in the rest of the tubes at a flowrate of 1 L/min gave the best performance. Comparing the best PV panel performance by the reference PV panel performance, increments in electrical and overall exergy efficiencies by ~ 48 and 52%, respectively, were obtained. (Abdelrazik et al., 2020) carried out a parametric numerical study on the hybrid cooling system for PV panels illustrated in Fig. 6. As shown in the figure, the layer just beneath the back surface of the panel was used for conserving uniform temperature distribution in the panel and composed of nano-graphene-enhanced PCM (paraffin wax) to enhance its thermal behavior. Behind this layer, there was a passageway for pure water flow. The effect of PCM layer thickness, solar concentration, atmospheric temperature, and flow rate of cooling fluid on PV temperature and performance were numerically investigated by the aid of MATLAB mathematical model. The results revealed that PV cooling and electrical efficiency were enhanced by reducing PCM layer thickness, solar concentration, and atmospheric temperature and increasing cooling fluid flowrate. By comparing PV panel performance under using this hybrid cooling system versus panel without any cooling system under different climatic conditions to assess the system effect over all-year conditions, it was showed that the proposed system with 10% nanoparticles has 6.9, and 22% higher electrical efficiency than

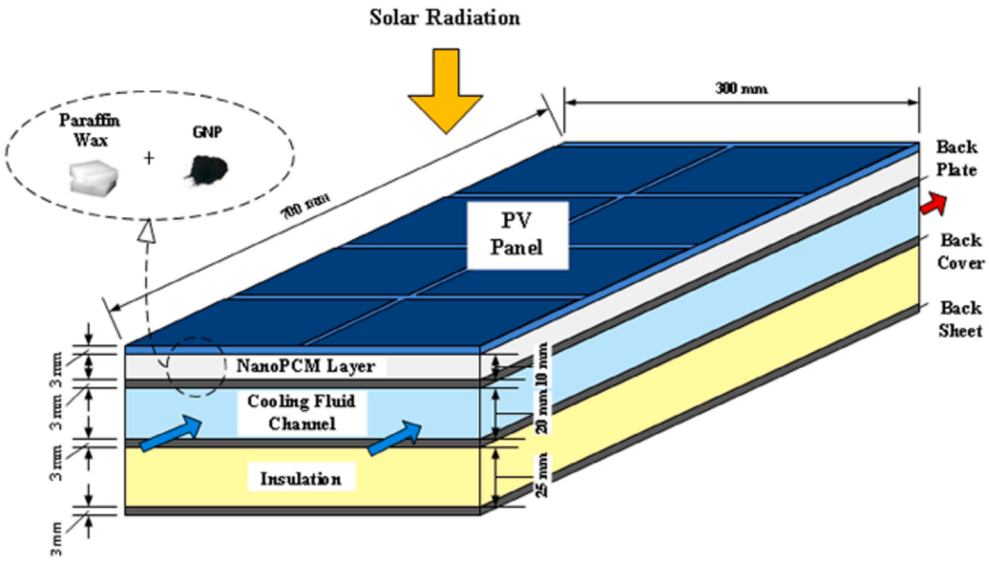


Fig. 6. Scheme of multilayer nano-based hybrid cooling system (Abdelrazik et al., 2020).

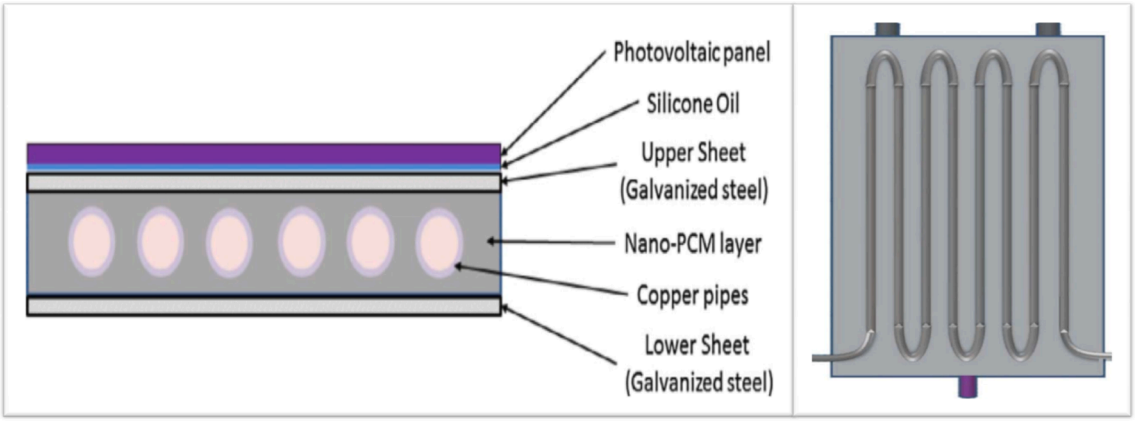


Fig. 7. Illustration for PV panel with direct contact hybrid nano-based cooling system: (a) cross-sectional view, and (b) top view (Al-Waeli et al., 2017b).

free PV panel in winter, and summer seasons, respectively.

(Al-Waeli et al., 2017b) proposed, validated, and tested a new hybrid cooling system for PV panels, shown in Fig. 7. Under the panel, a PCM container was installed with silicone oil to prevent the existence of any air bubbles, and copper tubes were located in the container for cooling NF flow. This PCM container was thermally insulated from the surroundings by means of 2 cm thickness glass wool. The container was equipped with two upper inlet ports: one for pouring the molten PCM and the other for venting. Besides, a lower port was for PCM drain process. To enhance the heat transfer process between PV panel and PCM, the PCM temperature was decreased by cooling fluid refreshment by means of a heat exchanger, as illustrated in Fig. 8. As shown, in the heat exchanger, cooling water was used to extract the NF heat, and the hot water was transported to a tank to store this energy. For study and validation purposes, four experimental study cases were carried out on a conventional PV panel, water-cooled PV panel, hybrid cooling system with pure paraffin wax PCM in the tank and water as a cooling fluid, and hybrid system with nano-SiC-enhanced paraffin wax PCM in the tank and SiC-water NF as a cooling fluid. Adding nanoparticles for PCM not only accelerated its melting and solidification processes but also helped it to maintain its properties after these processes. Electrical efficiency of 13.7% was recorded from applying the hybrid cooling system of N-PCM and NF. Besides, an increment in generated power by 96.75% compared to conventional PV panel was obtained.

Artificial neural network (ANN) is an algorithm that mimics processes and functions of the human brain on computer to analyze and predict the behavior of a set of functions and experimental results of a real system (Singh et al., 2001). For the previously illustrated four experimental study cases of the system proposed in (Al-Waeli et al., 2017b), (Al-Waeli et al., 2018) utilized the experimental results in prediction using three ANN models: SVM, MLP, and SOFM, in which PV current and efficiency were the outputs. The models' results showed that the PV panel cooled with N-PCM, and NF could record 13.32% electrical efficiency higher than that of conventional ones. It was noted that the results of SOFM and MLP models were similar to experimental results while SVM model results had less similarity than others, but its results also were located within an acceptable range. By using MATLAB program, (Al-Waeli et al., 2019a) investigated a mathematical model for the nano-based hybrid cooling system proposed in (Al-Waeli et al., 2017b) with N-PCM and NF. The comparison between experimental and mathematical model results showed the validation of the model because of a fair agreement. It was revealed that electrical and thermal efficiencies for experimental results were 13.7 and 72%, while for the mathematical model, they were 13.2 and 71.3%, respectively. On the other hand, as this upgrade requires an additional cost, so (Al-Waeli et al., 2019b) conducted a techno-economic study for this system with

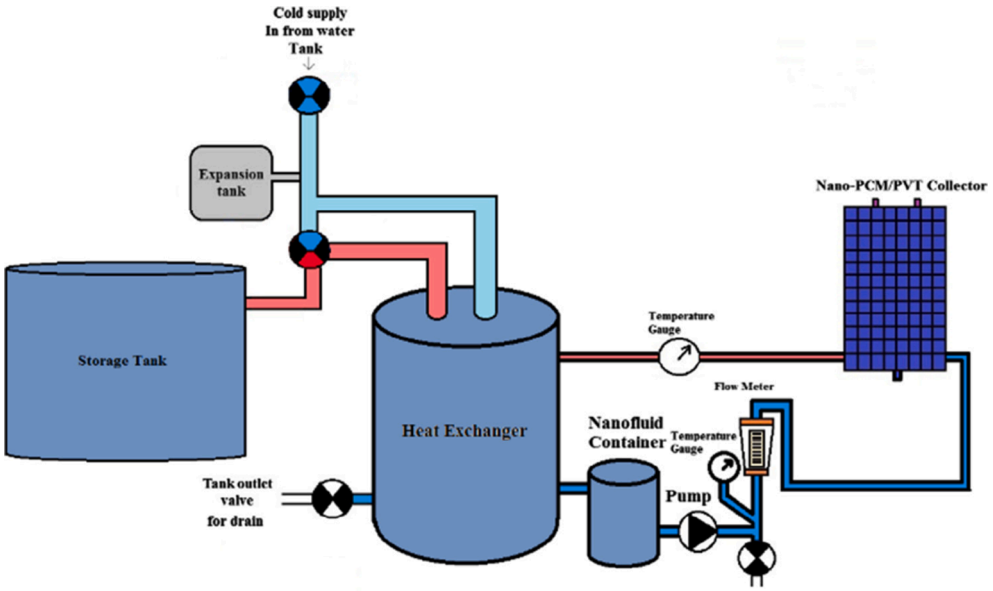


Fig. 8. Scheme for experimental setup for PV/T system with nano-based hybrid cooling system (Al-Waeli et al., 2017b).

the aid of MATLAB software program to evaluate its economic feasibility. It was revealed that the optimum NF flow rate was 0.175 kg/s and the best mass fraction of nanoparticles in PCM was 0.1%. Also, the results of the economic study showed that the proposed system was economically feasible and promising as it was able to produce electricity with a cost of 0.125 $/kWh with a payback period of 5–6 years. Recently, (Al-Waeli et al., 2020) proposed mathematical linear and ANN models to predict the power output from systems discussed in (Al-Waeli et al., 2017b). The prediction results were verified by experimental results and literature review. The results verified the validation and effectiveness of the proposed prediction methods. Also, a sensitivity study for factors affecting the system electrical and thermal outputs was conducted and showed that solar irradiance and electrical current had more impact than ambient temperature and voltage on systems' output.

(Hosseinzadeh et al., 2018) carried out a comprehensive study on the performance of conventional, NF-cooled, and hybrid nano-cooled PV panels. In NF-cooled PV panel, 0.2 wt% ZnO-water NF was circulated in a serpentine copper tube attached to the backside of the PV panel. Similarly, the hybrid nano-cooled PV panel had the same tubing system but was surrounded by paraffin wax PCM. The results illustrated that NF-cooled and hybrid nano-cooled PV panels had 7.1, and 11.9%, respectively, higher electrical power output compared to the conventional PV panel. The exergy analysis showed that the exergy efficiency of hybrid nano-cooled PV panel was higher than NF-cooled, and conventional PV panels by 10.02, and 26.84%, respectively. (Sardarabadi et al., 2017) experimentally investigated three PV panels: hybrid-cooled, fluid-cooled, and conventional. For the hybrid-cooled PV panel, a copper sheet and longitudinal circular tubes surrounded with PCM were integrated into the PV panel backside, while the fluid-cooled PV panel was equipped only with the copper sheet and tubes. The experimented cooling fluids were deionized water and 0.2 wt% ZnO-water NF, while PCM was pure paraffin wax. The results showed that PV panel temperature could be reduced by ~ 10 and 16 °C by applying the proposed fluid-cooling and hybrid-cooling systems, respectively. Using NF instead of deionized water had a slight enhancement in PV panel performance. Comparing with the conventional PV panel, the hybrid nano-cooled PV panel showed 13 and 23% increment in electrical power output and overall exergy efficiency, respectively. Depending on a similar cooling system design but with different cooling fluid and PCM, (Sarafraz et al., 2019) experimentally investigated the effect of nano-based hybrid cooling technique on PV panel performance. The experimented NFs were 0.1, 0.2 and 0.3 wt% multi-walled carbon nanotube-(1:1) water/ethylene glycol, while PCMs were 0.1, 0.2 and 0.3 wt% nano-multi-walled carbon nanotube -enhanced paraffin wax. The results illustrated that the hybrid-cooled PV panel with 0.3 wt% N-PCM and 0.2 wt % NF had the best performance as this PCM had the most enhanced thermal properties, and this NF attained enhanced thermal, rheological, and viscosity properties. Because of PV panel temperature reduction, this best study case showed ~ 20% enhancement in electrical power output, compared to the conventional PV panel.

To sum up, the most important findings of reviewed and discussed investigations conducted on nano-enhanced cooling techniques for PV panels are summarized in Table 1.

7. Conclusions and prospect recommendations

This paper aimed to merge the efforts conducted for enhancing the PV panels' performance via the utilization of nano-based cooling methodologies. The reviewed works were classified into separate or hybrid nano-based PV cooling units: nano-enhanced PCMs (N-PCMs), nanofluids (NFs), and hybrid units. Based on VOSviewer software, the research trend was analyzed. In addition, governing equations utilized for obtaining materials' properties with/without nanoparticles that help in PV or PV/T performance evaluation were driven. Also, detailed illustrations of designed systems were presented, and the main results were summarized. Based on the surveyed investigations, the following major points can be concluded:

- The PV panels' surface temperature can be reduced by up to 13 °C using N-PCMs.
- The higher nanoparticles' concentration in PCMs, the better PV temperature reduction can be obtained.
- For PV cooling via finned heat sinks immersed in N-PCMs, the higher the fins' number, the better PV performance can be achieved.
- Utilizing an NF-based PV cooling unit, the surface temperature reduction can reach up to 13.5 °C with conversion efficiency increasing ratio up to 22%.
- Integration nano-based hybrid cooling units can attain up to 16 °C surface temperature reduction and can enhance the electrical efficiency by up to 50%.

Based on this review, it can be noticed that PV cooling based on PCM

Table 1
Summary of conducted studies on PV panels cooling via N-PCM and NF.

Reference	Material	Most important findings
PV cooling via nano-enhanced PCMs (N-PCMs)		
(Sharma et al., 2017)	Nano-CuO-enhanced paraffin wax	11.2–12.5 °C reduction in PV panel temperature.
(Nada et al., 2018)	2% nano-Al_2O_3-enhanced RT_{55} paraffin wax	10.6 °C reduction in PV panel temperature and 13.2% increment in its efficiency.
(Abdelrahman et al., 2019)	0.1, 0.22, 0.33, 0.44, 0.55, 0.66 and 0.77 vol% nano-Al_2O_3-enhanced RT35HC PCM	36.9–52.3% reduction in PV panel temperature.
(Jamil et al., 2021)	Nano-enhanced PT-85 PCM with 0.25 and 0.5 wt% of multi-walled carbon nanotubes, graphene nanoplatelets and magnesium-	0.5 wt% graphene nanoplatelets-PT-85 PCM showed a 9.94 °C reduction in PV panel temperature and 11.11% increment in its efficiency.
Nanofluid (NF)-cooled PV panels		
(Sathyamurthy et al., 2021)	2% (1:1) carbon nanotubes/Al_2O_3-water NF	13.5 °C reduction in PV panel temperature and 21.73% increment in its electrical efficiency.
(Alous et al., 2019)	0.5 wt% multi-walled carbon nanotubes-water and graphene nanoplatelets-water nanofluids	Using 0.5 wt% graphene nanoplatelets-water nanofluid increases total energy and exergy efficiencies of the PV/T system by 63.1 and 20.6%, respectively.
(Abdallah et al., 2019)	0–0.3% multi-walled carbon nanotubes-water NF	Using 0.075% multi-walled carbon nanotubes-water NF increases the total energy efficiency of the PV/T system by 61.23%.
(Hussain and Kim, 2018)	Al_2O_3-water and CuO-water NFs	Using Al_2O_3-water and CuO-water NFs increases the electrical power output by 36.64 and 62.9%, respectively comparing with the water-cooled PV panel.
(Das et al., 2020)	0.2 wt% MXene (Ti_3C_2)-water + (IL), MXene (Ti_3C_2)-palm oil and Al_2O_3-water NFs	0.2 wt% MXene (Ti_3C_2)-water + (IL) reduces the panel temperature by 12.5 °C and increases its electrical efficiency by 13.95%.
(Hussain et al., 2019)	0–0.75% SiO_2-water, Al_2O_3-water, and CuO-water NFs	Using dual-fluid PVT with 0.75% CuO-water NF and water achieves 90.3% of total energy efficiency.
Hybrid cooling for PV panels		
(Abdollahi and Rahimi, 2020)	0.009 w/w Boehmite nano powder-coconut oil (82 wt%) + sunflower oil (18 wt %)	The proposed hybrid cooling system can enhance the PV panel efficiency by 48.23%.
(Nasef et al., 2019)	4% CuO-water NF	Using NF in the proposed hybrid cooling system instead of using pure water reduces the panel temperature by 4 °C and increases its efficiency by 2.7%.
(Salem et al., 2019)	0–1 wt% nano-Al_2O_3-enhanced $CaCl_2H_{12}O_6$ PCM	52% enhancement in panel overall exergy efficiency.
(Abdelrazik et al., 2020)	Nano-graphene-enhanced paraffin wax	Using the proposed hybrid cooling system increases the electrical efficiency of the PV panel by 6.9–22%, depending on the weather.
(Al-Waeli et al., 2017b)	Nano-SiC-enhanced paraffin wax and SiC-water NF	The proposed hybrid cooling system can achieve electrical efficiency of 13.7%.
(Hosseinzadeh et al., 2018)	0.2 wt% ZnO-water NF	11.9% higher electrical power output from the panel.
(Sardarabadi et al., 2017)	0.2 wt% ZnO-water NF	16 °C reduction in PV panel temperature
(Sarafraz et al., 2019)	0.1, 0.2 and 0.3 wt% multi-walled carbon nanotube-(1:1) water/ethylene glycol and 0.1, 0.2 and 0.3 wt% nano-multi-walled carbon nanotube-enhanced paraffin wax	Using 0.3 wt% N-PCM and 0.2 wt% NF increases PV panel electrical power output by ~ 20%.

and nanofluids is a promising and attractive research area. Hence, it is expected that more extensive relevant investigations and innovations will be conducted to develop nano-based PV cooling systems to enlarge the applicability of PV panels with high conversion efficiency. As a contribution to this trend, the following suggested investigations can be conducted:

- Influence of integrating hybrid nanomaterials, either in PCMs or nanofluids, on the performance of PV or PV/T systems.
- Studying the PV performance enhancement ratio obtained via using nano-coated heat sinks immersed in pure/nano-enhanced PCMs.
- Evaluating the PV performance under air-based cooling in the presence of different turbulators containing nano-enhanced PCMs.
- Effect of hybrid PCMs (organic/synthesized) with/without nanoparticles on the performance of PV panels.
- Numerical modeling of using nano-based hybrid cooling (with different configurations) in PV cooling for grid-connected large-scale plants: techno-economic assessment.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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